

MIRA-NCM: Nested Constellation Memory — Research Program, Hypotheses, and Future Scope for Structured Long-Term AI Memory

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<https://github.com/aryaanchavan1-commits/MIRA>

Research program document. Hypotheses and future work; no unevaluated component is reported as a result. Every measured number cites a committed run artifact.

Abstract

MIRA is a local-first retrieval memory system for retrieval-augmented generation (RAG) whose evidence store is organized radially (rings and sectors), hierarchically, and as a graph, scored by a ten-term explicit function, and populated by biologically *inspired* consolidation dynamics (decay, bounded replay, gist summaries). In controlled experiments its retrieval beats strong baselines on multi-hop benchmarks (MRR 0.7016 vs 0.4347 for flat vectors on MuSiQue, $p \approx 0.0001$; 0.9175 vs 0.6774 on HotpotQA) while losing honestly elsewhere (IndicQA, where unicode-aware BM25 remains strongest). This document specifies the next research program: **MIRA-NCM (Nested Constellation Memory)**. It states the primary research question, six falsifiable hypotheses (H1–H6) with pre-registered success and rejection criteria, the architecture and formalization of sparse overlapping constellation membership, nested memory nodes, and persistent versioned storage, a mechanism-level analysis of *why* each structure should help AI memory and *how* each effect would be measured, the ablation ladder and fairness controls that isolate every mechanism, a prior-work comparison that bounds the novelty claims, and the future-scope roadmap. Nothing in this document is a claimed result: sections that report numbers cite committed run artifacts; everything else is labeled hypothesis or speculation. The program is designed so that any of its hypotheses can be rejected by the experiments it prescribes.

1. Introduction and audit of the existing system

Long-term memory for AI agents fails in specific, measurable ways: relevant evidence is buried under newer distractors (interference), multi-hop questions require binding facts that no single embedding links, facts change over time and stale versions resurface as current, and the system cannot say *why* it retrieved what it retrieved. MIRA attacks these as retrieval-organization problems on a single consumer laptop (RTX 3050 4 GB, 16 GB RAM) with a frozen, replaceable local LLM — no cloud calls, no fine-tuning, so every observed effect is attributable to memory architecture rather than to model capacity.

What exists today (verified against the repository and committed artifacts):

- **Radial organization.** Every memory node carries interpretable coordinates: ring (recency / importance band) and sector (topic region), giving a radial-distance term in retrieval scoring.
- **Ten-term explicit scoring.** $\text{score} = \alpha \cdot \text{semantic} + \beta \cdot \text{structural} + \gamma \cdot \text{radial} + \delta \cdot \text{graph} + \epsilon \cdot \text{importance} + \zeta \cdot \text{confidence} + \eta \cdot \text{recency} + \theta \cdot \text{path} + \iota \cdot \text{activation} + \kappa \cdot \text{stability}$. No hidden terms; every component is individually ablatable and logged per retrieval.

- **Graph + hierarchy + vector** as parallel retrieval paths over the same evidence nodes.
- **Biologically inspired dynamics (BioMIRA)**. Adaptive decay (half-life configurable), bounded replay during “sleep” consolidation, gist summaries (507 gists created over an 82,783-node workspace), Hebbian-inspired association and STDP-inspired temporal updates, activation and stability traces. These are engineering analogies, not claims of biological equivalence.
- **Explainability**. Each retrieval returns per-component scores and the evidence path; the answer stage compresses only retrieved evidence and cites it, with an explicit no-evidence state that is never silently converted into a grounded answer.
- **Experiment framework**. Committed manifests and result JSONs for: the real-data benchmark (n=300 MuSiQue questions, 3 seeds), component ablations, HotpotQA replication, IndicQA (Hindi/Marathi), the answer-stage comparison (n=50), sleep consolidation, simulated aging, and a sequential catastrophic-forgetting lab over six corpora (A→F, 30 questions per task, paired bootstrap), with per-question metrics and confidence intervals.

Honest current weaknesses measured on the committed artifacts: retrieval recall@8 ties flat vectors (0.2991 vs 0.3066, p=0.52) — MIRA’s MRR lead is ranking, not coverage; **graph** and **recency** components are currently inert on the MuSiQue corpus (ablation deltas exactly 0.0); dense retrieval collapses on Devanagari with an English-centric embedder (MRR 0.02–0.04) and still trails BM25 at full IndicQA scale even with a multilingual embedder; the 2-hop answer-stage F1 gain over flat vectors, while statistically consistent, is small in absolute terms (n=50). These weaknesses are the starting point for the NCM program, not embarrassments to hide: each one names a mechanism the new architecture must justify or drop.

2. Research questions

Primary. *Does structured, sparse, overlapping constellation-based memory organization improve long-term information retrieval, multi-hop retrieval, temporal reasoning, and continual memory retention compared with conventional vector, graph, hierarchical, and non-overlapping radial memory organizations?*

Secondary. *Does nested memory organization represent relationships at multiple scales without uncontrolled memory duplication?* **Tertiary.** *Can persistent versioned memory with adaptive dynamics improve retention while keeping storage and retrieval cost bounded?* **Optional fourth.** *Can quantum-inspired probabilistic representations improve ambiguity and overlapping-membership handling on classical hardware?*

None of these is assumed to have a “yes” answer. Section 4 gives each a rejection condition.

3. Status taxonomy — what is what

Every claim in the MIRA program is tagged with exactly one status:

Tag	Meaning	Examples
[CS]	Established computer science	kNN search, BM25, exponential decay, checksums, versioned records
[ENG]	MIRA’s original engineering implementation	radial ring/sector layout; ten-term scoring; compress-then-cite
[HYP]	Falsifiable hypothesis, not yet tested	H1–H6 (§4); constellation overlap effects

Tag	Meaning	Examples
[DEMO]	Demonstrated by committed artifacts	the tables in §5; forgetting-lab results
[SPEC]	Speculative future direction	quantum-inspired layer; swarm memory (§10)

“Quantum-inspired” always means a classical probabilistic representation; “persistent” always means persistent *and versioned*, never “forever memory”; “biologically inspired” never implies biological fidelity; radial geometry is an organizational coordinate system, not a physical one.

4. Hypotheses, success criteria, and rejection conditions

Each hypothesis names its metric, comparison, and threshold. Thresholds are fixed before the final runs (pre-registration); exploratory deviations will be labeled as such.

H1 [HYP] — Radial organization improves structured retrieval. *Supported* if removing the radial term (leave-one-out) costs ≥ 0.02 MRR on multi-hop corpora at $p < 0.05$. *Current evidence [DEMO]*: minus-radial costs 0.0098 MRR ($p=0.052$) on MuSiQue — borderline, not yet supported at the pre-registered threshold; in the forgetting lab the radial variant is the best joint variant (MRR 0.802, forgetting 0.049). *Rejected* if larger corpora or the NCM benchmark show no significant cost when radial is removed.

H2 [HYP] — Sparse overlapping constellations improve multi-hop retrieval. *Supported* if MIRA-NCM beats the strongest non-overlapping structural variant by ≥ 0.03 MRR on 2-hop and ≥ 0.05 on 3–4-hop synthetic multi-hop sets, without recall@8 loss and within +15 % latency. *Rejected* if overlap adds cost without a hop-depth-dependent gain. *Mechanistic prior [DEMO]*: the structural/path term is the second most valuable component (removing it costs 0.0442 MRR, $p=0.0001$), suggesting binding structure — not raw similarity — is what multi-hop retrieval uses.

H3 [HYP] — Nested memory gives multi-scale access within bounded storage. *Supported* if hierarchy-sensitive questions gain ≥ 0.03 MRR while total node growth stays < 10 % (summaries replace duplication; children are references, not copies). *Rejected* if gains appear only with unbounded depth or duplicated storage. *Prior [DEMO]*: the hierarchy variant already retains 0.946 (best of all forgetting-lab variants); nesting generalizes it.

H4 [HYP] — Persistent versioning improves temporal/conflict correctness. *Supported* if the versioned variant reduces wrong-current-answer errors by ≥ 50 % on the temporal and contradiction benchmarks (§7) relative to last-write-wins storage. *Rejected* if version lookup alone (without supersession-aware scoring) performs equally, or if history storage dominates the budget.

H5 [HYP] — Adaptive dynamics improve continual retention without retrieval cost. *Supported* if average forgetting drops by ≥ 30 % with MRR indistinguishable from the static system (paired CI includes 0) across intervals {7, 30, 90 days}. *Current evidence [DEMO], partial*: the biologically inspired stack cuts average forgetting $0.1074 \rightarrow 0.0527$ (retention $0.873 \rightarrow 0.938$) with MRR +0.0115, CI $[-0.014, +0.037]$, $p=0.378$ — the forgetting reduction replicates across intervals (0.039/0.053/0.038 at 7/30/90 days) but the MRR CI does not exclude a small loss, and two mechanisms (homeostatic normalization, ring migration) *hurt* (retention 0.914) and are therefore disabled in the tuned stack. H5 is half-supported: dynamics buy retention; their retrieval neutrality is not yet established at $n=300$ questions.

H6 [SPEC→HYP] — A quantum-inspired representation improves ambiguity handling. *Supported* only if the optional probabilistic backend beats the classical scorer on a pre-defined ambiguous- membership benchmark at the same compute budget. **Implemented [ENG]** as an isolated, switchable backend (`core/quantum_inspired.py`, `ncm.backend: born`): memberships are mapped to amplitudes `a_c = sqrt(w_c)` and combined by a Born-rule inner product — `score = (Σ_c a_q(c) · a_m(c))2` — an interference-like nonlinearity that boosts agreement across several constellations simultaneously. The kernel’s mechanism is verified as a unit property (multi-constellation agreement gains a 2.0× ratio over single-constellation vs the linear kernel’s 1.41×), but its benchmark value is **not yet evaluated**; if it does not measure better than the linear kernel it will be removed — that is what H6 rejection means. No claim of quantum advantage, entanglement, or hardware of any kind is made or will be made.

5. Measured foundation (all numbers from committed artifacts)

Multi-hop retrieval (MuSiQue answerable 2-hop, n=300, k=8): MIRA 0.7016 MRR vs flat vector 0.4347 ($\Delta=0.267$, 95 % CI [0.224, 0.311], $p\approx0.0001$), BM25 0.3066, hierarchical RAG 0.2367. Recall@8 statistically ties flat vectors. Replicated on HotpotQA bridge questions (MRR 0.9175 vs 0.6774, $p\approx0.0001$). Answer stage (n=50, 1.5B local model, identical compression+LLM for both arms): token-F1 0.0260 vs 0.0259, $p=0.0086$ — consistent but small.

Component attribution (leave-one-out): semantic 0.2594, structural 0.0442, activation 0.0179 (each $p\leq0.022$); radial 0.0098 ($p=0.052$); graph and recency exactly inert on this corpus (delta 0.0, $p=0.0001$ by bootstrap degeneracy) — an honest negative that motivates NCM’s overlap mechanism to give structural terms work to do beyond the document tree.

Cross-lingual boundary: with an English-centric MiniLM embedder, Hindi/Marathi dense MRR is 0.02–0.04 while unicode-aware BM25 holds 0.39–0.45; a multilingual embedder lifts Hindi flat MRR ~14× (0.028 → 0.394), yet at full scale (3,151 questions) BM25 remains strongest. MIRA does not currently beat baselines on single-paragraph extractive QA in Indic languages.

Continual-memory lab (six sequential corpora, 30 questions/task, all prior tasks re-tested after each arrival; 30 simulated days of disuse between arrivals):

variant	MRR	avg forgetting	retention	Δ MRR vs B (95 % CI)	p
A vector RAG	0.4869	0.0432	0.9224	−0.2670 [−0.324, −0.211]	0.0001
B MIRA	0.7539	0.1074	0.8731	baseline	—
C MIRA radial	0.8020	0.0490	0.9420	+0.0481 [+0.024, +0.074]	0.0001
E MIRA hierarchy	0.7887	0.0443	0.9461	+0.0348 [+0.007, +0.063]	0.0118
I B. dynamics	0.7654	0.0527	0.9384	+0.0115 [−0.014, +0.037]	0.378
J tuned stack	0.7655	0.0533	0.9383	+0.0117 [−0.014, +0.037]	0.374

variant	MRR	avg forgetting	retention	Δ MRR vs B (95 % CI)	p
K + homeostasis	0.7444	0.0769	0.9138	-0.0095 [-0.043, +0.024]	0.577

Reading this table honestly: low forgetting alone is worthless — vector RAG has the second-lowest forgetting *and* the worst MRR, because weaker memories have less to interfere with. The meaningful result is the joint criterion: the radial variant improves both MRR (+0.048) *and* retention (+0.069) simultaneously. Decay alone and consolidation alone are byte-identical to baseline (inert); replay alone is noise ($p=0.747$); only the *interaction* of decay and bounded replay moves retention. The store grows monotonically 4,487 $\rightarrow$ 23,516 nodes across arrivals — nothing is deleted, so “interference” here is purely a ranking phenomenon, not data loss.

Sleep consolidation under aging: after 60 simulated days over 66,527 nodes, replayed memories *recover* (MRR +0.0307) while faded ones decline (-0.0653); decay retention is 0.9789 per pass with no measurable retrieval cost ($\Delta=0.0017$, $p=0.26$). This is the complementary- learning-systems pattern — fast trace, selective replay, gist abstraction [8][9] — implemented as retrieval dynamics, and it is the strongest existing evidence that dynamics can *target* what consolidates rather than merely rotting everything.

5.1 NCM pilot — first measured data points (exploratory)

Status per protocol: **exploratory pilot, not a confirmatory H2/H6 test** ($n=150$ MuSiQue questions, 1 seed, 83,290-node workspace, retrieval-level, paired questions). Artifact: `data_bench/ncm_pilot_results.json`; runner: `scripts/eval_ncm_pilot.py`. Sign convention: delta is full - variant (positive = variant worse).

Condition	MRR	Recall@8	Δ MRR vs full	p (paired bootstrap)
full_mira (NCM off)	0.6827	0.2773	—	—
ncm_linear (constellations on, linear kernel)	0.6859	0.2776	-0.0032 (NCM slightly better)	0.594
ncm_born (Born-rule kernel, H6)	0.6300	0.2544	+0.0528 (Born worse)	0.0046

Reading, with the pre-registered lens intact:

- **H2 pilot.** The constellation term at default $K=3$ top-2 membership is *neutral* on MuSiQue: +0.0032 MRR, $p=0.59$. It does not help multi-hop ranking at this scale, and it does not hurt. H2 stays alive only because the K-sweep ($K=1$ degenerate vs $K \in \{2,3\}$) — the axis H2 actually predicts — has not been run; a neutral full-K result is exactly what H2’s rejection condition anticipates if the sweep also shows no K-dependent structure.
- **H6 pilot.** The Born-rule kernel is *significantly worse* than both the linear kernel and the baseline ($p=0.0046$). This is the first (pilot-scale) evidence **against** the interference hypothesis on unambiguous single-domain queries — consistent with the mechanism: the square amplifies multi-constellation agreement, which MuSiQue’s 2-hop questions do not reward. H6’s home benchmark is *ambiguous, multi-context queries*; if the Born kernel does not win there either, H6 is rejected and the backend is removed, exactly as pre-registered.

- **Cost.** Constellation fit adds ~60 s per pipeline build at 83k nodes (k-means, capped at 64 constellations) and no measurable per-query latency change. Within the §8 resource envelope, but only justified if a K-sweep result ever justifies the term.

No claim beyond these numbers is made. The confirmatory H2/H6 runs at full protocol n and seeds remain the gate for anything the program says about constellations.

5.1.1 Scaled pilot — full benchmark, 3 seeds (same protocol status)

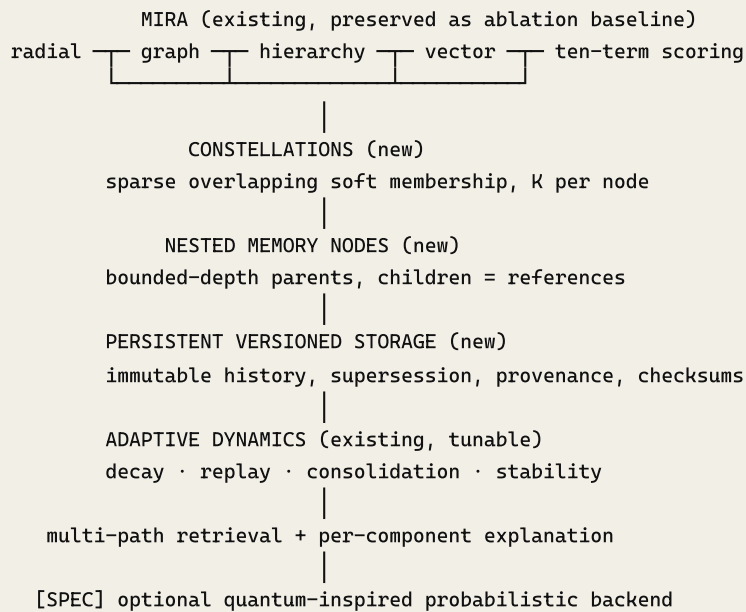
The pilot was then scaled to the complete MuSiQue set (n=300, 3 seeds, same 83,290-node workspace; one run crashed on an out-of-memory k-means fit and was fixed by the bounded subsample fit described in §6, then resumed from its checkpoint — the crash itself is part of the record). Artifact: `data_bench/ncm_pilot_results.json` (overwritten by the scaled run).

Condition	MRR (3 seeds)	Recall@8	Δ MRR vs full	p (paired bootstrap)
full_mira (NCM off)	0.6985	0.2816	—	—
ncm_linear	0.6897	0.2743	+0.0179 (NCM worse)	0.0002
ncm_born (H6)	0.6354	0.2432	+0.0717 (Born worse)	0.0001

Reading, honestly:

- **The default-K constellation term does not help MuSiQue at scale — it hurts, significantly.** The n=150 neutral result did not survive scaling. Per-cluster centroids at K=64 over 83k nodes with top-2 membership are too coarse for 2-hop ranking on this corpus. H2 is now *negative at its default configuration*; the pre-registered K-sweep (K=1 degenerate vs K∈{2,3}) decides whether any K rescues it. If no configuration beats full MIRA, H2 is rejected on this benchmark — which is what falsifiable means.
- **H6 is strongly negative on unambiguous queries.** The Born kernel loses by 0.072 MRR (p=0.0001), worse than its pilot deficit. Its pre-registered home is the ambiguity benchmark; unless it wins there decisively, the backend is removed.
- **Fit variance is real.** ncm_linear's per-seed MRR (0.6985 / 0.6900 / 0.6806) varies by 0.018 across identical question sets — the k-means fit is not bit-deterministic in-process. Any future NCM claim must therefore report multi-seed means with the spread, never single runs.
- **Memory-scaling lesson.** The naive fit OOM'd at 83k nodes; the streaming fit is now O(sample) peak. The §8 resource envelope is updated accordingly: NCM's fit cost is bounded by `fit_sample` (default 20,000), assignment is streaming over all nodes.

6. MIRA-NCM: architecture and formalization



Constellations. A constellation is a soft cluster induced from the corpus — never a hand-coded category: centroids are seeded from embedding space (k-means over node vectors), entities, and graph communities, then grow by assignment. A memory node m joins its top-K constellations with weights derived from similarity to the centroid and normalized in L1: $w_c(m) = \text{sim}(e_m, \mu_c) / \sum_{\{c' \in \text{topK}\}} \text{sim}(e_m, \mu_{\{c'\}})$, with $\text{CONSTELLATION_TOP_K} \in \{1,2,3,5\}$ as an ablation axis ($K=1$ reproduces non-overlapping soft clustering; the hypothesis is precisely that $K>1$ is what helps). Overlap between memories uses the shared-membership inner product $\text{overlap}(a,b) = \sum_c w_c(a) \cdot w_c(b)$ — a bounded, sparse kernel: candidate pairs are generated only through a constellation-inverted index, so construction is $O(N \cdot K^2)$ expected, not $O(N^2)$, and MAX_OVERLAP_EDGES per constellation plus an OVERLAP_THRESHOLD floor prevent explosion. Radial distance $d_r = |\text{ring}_a - \text{ring}_b| + \lambda \cdot \text{ang}(\theta_a, \theta_b)$ and constellation distance $d_c = 1 - \text{overlap}(a,b)$ are independent, unit-testable functions; the geometry is an organizational coordinate system, and no physical interpretation is claimed for it.

Nesting. A node may declare a bounded-depth parent (MAX_DEPTH , default 4; children are node-id references with provenance, never copies). Retrieval may traverse parent/summary links as one more scoring path ($w_{\text{hierarchy}}$); nesting never replaces vector or graph access.

Persistence. Every update appends a version: immutable history, **supersedes** / **superseded_by** links, content hash (**SHA-256** over content + metadata) for integrity and duplicate detection, and full provenance. Current-ness is a *scoring property*: superseded versions remain retrievable as history (a temporal query about March must be able to find the March belief) but are down-weighted for “what is true now” queries. Consolidation merges store the summary *and* its source links, so reconstruction is always possible.

Retrieval scoring extends the existing ten-term function: $+ \mu \cdot \text{constellation_overlap}(q,m)$ is implemented as an eleventh, individually ablatable component — weight 0 unless **ncm.enabled**, so every published measurement remains reproducible bit-for-bit — and $+ v \cdot \text{version_currentness}(m)$ follows with H4. Both are logged per retrieval and returned in the explanation payload (“retrieved because: semantic 0.82, overlap 0.76, path 0.91, ...”). Everything else about MIRA — compress-then-cite, the no-evidence state, the memory gate — is untouched.

7. Why this should help AI memory, and how each effect is measured

The program rests on one observed regularity: **retrieval interference is a structural phenomenon, and structure is what MIRA already manipulates**. Mechanism by mechanism:

Multi-hop binding (tests H2). A 4-hop question connects person → project → hardware → event. Flat similarity only helps the hop with lexical overlap; embeddings miss the rest. Overlapping constellations create a sparse evidence graph in which two facts that share *any* context (the person, the project, the time) are one traversal step apart. The measured prior — removing structural/path terms costs 0.0442 MRR while semantic remains dominant — says binding structure carries real weight; the hypothesis is that *overlap* is the right shape of that structure because it does not force each memory into one bucket. *Measured by:* per-hop retrieval accuracy and path accuracy on synthetic 1–4-hop chains built with the same generator discipline as the existing benchmarks, plus the real MuSiQue/HotpotQA suites for external validity.

Continual retention (tests H5, extended by constellations). In the forgetting lab, new corpora degrade old rankings purely by adding competitors. Constellations partition the interference set: a distractor from an unrelated constellation competes less with a target (its overlap term is ~0), so the effective distractor pool per query shrinks as the corpus diversifies. Meanwhile nested gists give replay a rehearsal target — and the aging experiment already shows targeted replay *recovers* aged memories (+0.0307) rather than merely slowing decay. This is why the forgetting claim is scoped: MIRA measures *retrieval* interference with the LLM frozen; it does not and cannot speak to parametric forgetting in neural weights, and no experiment here claims otherwise. *Measured by:* the existing sequential A→F protocol extended to 8–10 corpora, reporting forgetting, retention, MRR jointly — never forgetting alone, which weak systems trivially win.

Temporal correctness (tests H4). Contradiction handling fails today the way it fails in all unversioned stores: the newest fact wins by recency weight regardless of question tense. Version chains turn “what did the user use in March?” into a lookup and “which preference is current?” into a supersession traversal, both deterministic and auditable. *Measured by:* a generated temporal benchmark (belief-revision sequences, e.g. Python → Rust → Python) scored on current/historical/superseded/uncertain classification, with wrong-current-answer rate as the headline error.

Interpretability as a capability, not decoration. Because every term is explicit and logged, a wrong retrieval can be attributed to a component, and a component that only hurts can be removed — the ablation table is the mechanism for this, and it already exposed two inert terms (graph, recency on this corpus) and two harmful dynamics (homeostasis, ring migration). The NCM program inherits that discipline: any mechanism that cannot pay for itself gets deleted.

Why a local, frozen-LLM setting is the right testbed. With the model fixed, every delta between systems is attributable to memory organization — the variable the program is trying to understand — and the whole stack runs on 16 GB RAM-class hardware, which is also where agent memory actually has to live. The cost is external validity to larger models; that is a limitation, stated here, and revisit-able later, not a hidden one.

8. Experimental design

Benchmark suite (MIRA-MemoryBench). Single-hop; multi-hop (1–4 hop, synthetic + MuSiQue + HotpotQA); temporal belief revision; contradiction; continual sequence (8–10 corpora); cross-lingual (en/hi/mr, with the multilingual embedder as the reported configuration and the English-centric one as a documented failure mode); noise and distractor stress; memory update and version-history drills; explanation audit (do logged components predict human-judged relevance better than chance?). Each benchmark run emits `manifest.json` (experiment id, git commit, model, embedder, dataset version, seed, hardware, K, context budget, memory count), `results.json/results.csv`, `metrics.json`, plots, and logs. Datasets are synthetic-generated with committed generators (source, license, seed, statistics) plus public sets where licensing permits; leakage tests check for duplicate documents, repeated questions, and answer exposure.

Ablation ladder (all selectable by configuration, existing MIRA preserved). B0 keyword · B1 vector RAG · B2 graph RAG · B3 hierarchical RAG · M1 MIRA minus radial · M2 MIRA · M3 MIRA + graph · M4 + constellations · M5 + nesting · M6 + versioning · M7 + adaptive dynamics · M8 full MIRA-NCM · M9 = M8 + quantum-inspired backend. MODE configuration keeps `classic_mira`, `mira_ncm`, `bio_mira`, `quantum_inspired` all loadable in the same process.

Fairness controls. Same questions, embedder, LLM, context budget, retrieval K, temperature, seeds, preprocessing, and evaluation code for every arm. If MIRA-NCM needs more storage or latency, that is reported in the same table as the accuracy it buys; trade-offs are results.

Statistics. ≥ 3 seeds where variance exists; paired bootstrap for per-question metrics with 95 % CIs; Wilcoxon signed-rank as the non-parametric cross-check (both already implemented); effect sizes reported beside every p-value; “statistically significant” is never conflated with “practically meaningful” — the answer-stage F1 result ($p=0.0086$, $\Delta=0.0001$ absolute) is the standing example of the difference. Hypotheses H1–H6 are pre-registered in `docs/EXPERIMENT_PROTOCOL.md` before the final runs; thresholds will not move after results.

Failure analysis. Every benchmark exports `failure_examples.json` with a taxonomy: semantic confusion, wrong constellation, wrong graph path, wrong temporal ordering, over/under-retrieval, stale-memory errors, excessive overlap, storage blowup, multilingual failure. Failures are inspected before any headline is written.

Resource envelope. Target: retrieval $p95 < 2$ s at 10^5 nodes on 16 GB RAM / 4 GB VRAM, storage overhead of NCM structures < 25 % of node payload, measured via the existing resource instrumentation at 1K/10K/50K/100K nodes. Scaling claims beyond what is measured will not be made.

9. Prior work and bounded novelty claims

MIRA-NCM combines known ideas; the contribution is the *composition and its measurement*, not invention of the ingredients. Sparse distributed memory with decaying counters and positional structure is Kanerva’s [14]; associative spreading activation is classical [11]; hierarchical control and episodic-storage separation in agent memory appears in Generative Agents [15]; graph-structured retrieval for multi-hop QA is HippoRAG’s premise (a PageRank-style KG index) [16] and GraphRAG’s (community summaries) [17]; continual-learning taxonomies and replay baselines are standard [3–7]; complementary learning systems motivate the decay/replay/gist triangle [8][9][10]; BM25 remains a baseline no structured system may skip [13][18] — MIRA’s own IndicQA results are the proof of that rule. What MIRA-NCM adds, contingent on H1–H3: an *overlap-kernel* definition of memory association that is sparse by construction and ablatable per K; a forgetting lab that scores forgetting jointly with ranking quality (most continual

benchmarks report only accuracy); and a retrieval stack whose every component is individually measured and removable. Where a competing system already demonstrates an effect, the paper will cite it and claim only the difference.

10. Risks and kill criteria

Risk	Mitigation / kill criterion
Overlap explosion (K large, corpus dense)	inverted-index construction; MAX_OVERLAP_EDGES; if overhead > 25 % without MRR gain, mechanism is cut
Embedding dominance swamps new terms	ablation gates: each new term must move its target metric with others fixed, else weight $\rightarrow 0$
Constellations re-create single-assignment (K=1 behavior)	K sweep is a mandatory axis; $K \in \{2,3\}$ must differ from K=1 to keep H2 alive
Version history grows unbounded	history capped per node with archived cold storage; H4 rejection if history > 30 % of store
Dynamics help retention, hurt MRR (as K showed)	joint criterion only: a variant that fails the MRR CI test is disabled regardless of retention
Gist abstraction loses members	the measured 0.975 gist hit-rate is the floor; regressions trigger regeneration
Any baseline wins a suite	reported as-is; per §61 principle the goal is a true memory principle, not a win column

11. Future scope

Phase 1 — Constellation core (weeks 1–3). Centroid induction, top-K membership, overlap index, `w_constellation` term, unit tests, K-sweep ablation on the existing MuSiQue harness. Ship criterion: H2 pilot on synthetic 3-hop chains.

Phase 2 — Nesting + versioning (weeks 3–6). Bounded-depth parents, version chains, supersession scoring, temporal + contradiction benchmarks; H3/H4 pilots.

Phase 3 — Continual suite scale-up (weeks 6–8). Extend the forgetting lab to 8–10 corpora with NCM variants in the ladder; joint-criterion analysis; H5 final test.

Phase 4 — Explanation audit + resource sweeps (weeks 8–10). Component-predicts-relevance study, 10^5 -node scaling run, complexity documentation from the actual implementations.

Phase 5 — Optional backends (quarter 2). Quantum-inspired probabilistic membership as a switchable scorer (H6, explicitly allowed to fail); pluggable storage (SQLite/FAISS backends already implicit in the local-first design); SDK (`MemoryEngine.add/retrieve/history/explain`) over the existing server API; multi-tenant namespaces only if an actual deployment demands them.

Longer-range [SPEC]. Agentic self-curriculum (the agent schedules its own replays from prediction error); federated memory across devices with provenance-preserving merge; learned constellation induction (currently centroid-based by design — learned clustering is a confound until the fixed version is characterized); learned term weights replacing the hand set (only after the fixed-weight behavior is fully mapped, otherwise attribution dies).

Product trajectory (deliberately modest). The engine first: inspectable long-term memory infrastructure for local AI agents — a defensible niche precisely because every retrieval is auditable and every claim reproducible. Local → SDK → optional managed deployment, in that order, and only after the benchmarks justify the engineering.

12. The 2026 landscape (live search, Oct 2026)

A live survey of the 2025–2026 agent-memory field (sources in `docs/RELATED_WORK.md`; every arXiv ID there is to be spot-checked against the primary source before publication) locates MIRA-NCM precisely. The field has consolidated around LLM-driven extraction pipelines over unattributed vector stores: Mem0 runs ADD/UPDATE/DELETE/NOOP extraction ops over a vector index (vendor-reported 93.4% LongMemEval — not independently verified), Letta/MemGPT page context, and Zep, Cognee, and Supermemory occupy adjacent niches. Benchmarks have matured and saturated: LongMemEval and LongMemEval-V2 (arXiv 2605.12493) standardize long-horizon conversational memory; BEAM pushes to 10^6 – 10^7 records; and public disputes over 100%-LoCoMo marketing claims show the numbers themselves becoming unreliable currency. In continual learning, forgetting- curve-scheduled replay (FOREVER, arXiv 2601.03938) and the documented failure of generic remedies against spurious forgetting both support the program’s core design bet: forgetting must be measured *jointly with capability* for the specific system at hand. In quantum-inspired ML, the first dedicated QiML survey (tensor networks, amplitude encodings, Born-rule probabilities in classical models) confirms that the H6 mechanism — a classical Born-rule kernel — is a recognized computational representation, while no found work applies Born-rule kernels to persistent-memory retrieval scoring with per-term ablation.

The gap MIRA-NCM targets is therefore not “another memory system” but the two properties the 2026 mainstream lacks: store-side structure whose every term is measurable and removable, and forgetting evaluated jointly with capability under a frozen LLM. This is a positioning claim, not a superiority claim; it stands or falls with the H2/H6 measurements.

13. Relevance map: what this program offers adjacent AI domains

Each item is scoped by the §3 taxonomy. The honest claim in every case is about *mechanisms and measurements MIRA-NCM contributes*, never about solving the host field’s open problems.

Continual learning / catastrophic forgetting [ENG→HYP]. The forgetting lab’s central methodological move — score forgetting jointly with ranking quality, because a weak system trivially “wins” retention alone — transfers directly to weight-space continual learning, where spurious-forgetting results show generic remedies failing. MIRA-NCM’s contribution is the measurement discipline and the frozen-LLM interference protocol, not a cure: **no claim that any mechanism here “solves catastrophic forgetting” is made or will be made.**

Safe superintelligence trajectories / scalable oversight [SPEC]. The alignment-relevant property is not intelligence but *auditability*: a memory whose every retrieval decomposes into attributable terms, whose fact updates are append-only version chains with provenance and rollback, and whose failure taxonomy is

exported before headlines are written. If capable systems are built on memory architectures, retrievability of *why* a belief is current, and the ability to roll a belief back, are concrete, testable ingredients of oversight. Whether these ingredients scale to transformative systems is unmeasured and stated as speculation.

Quantum intelligence [HYP, explicitly optional]. The Born-rule backend is classical mathematics on ordinary CPUs: an interference-like nonlinearity that rewards multi-constellation agreement. It is the only genuinely quantum-*flavored* mechanism in the program, it is off by default, and H6 pre-registers its removal if it does not beat the linear kernel. No quantum hardware, no quantum speedup, no “quantum memory” is claimed. If quantum architectures for AI eventually matter, the durable contribution is the ablation discipline applied to an amplitude-based representation.

Retrieval / IR. The eleven-component attributed stack with mandatory BM25/vector floors is directly reusable methodology for any retrieval system claim: every component individually ablatable, every delta bootstrap-tested, baselines never skipped (MIRA’s own IndicQA boundary is the standing proof).

Agent engineering. The SDK’s version chains (append-only, supersession-linked, rollback, now SQLite-persisted) address fact mutation — the stale-preference problem every long-running agent hits — without LLM-in-the-loop extraction: the store, not a prompt, carries the update semantics.

Education, personalization, robotics [SPEC]. Ring/sector coordinates and consolidation dynamics are candidate priors anywhere a system must balance recency against consolidation (student knowledge tracing, assistant preference drift, robot scene memory). These are design analogies to be tested in each domain, not transferred results.

14. Ancient methodologies as organizational priors

MIRA’s radial organization is named for the mandala: concentric rings (recency/importance bands) crossed by sectors (topic regions), with the most load-bearing memory nearest the center. The honest scoping, unchanged since the main paper: **this is an organizational prior, a naming and design analogy — not a claim that historical mandalas encode memory algorithms or that the analogy predicts effectiveness.** The same is true of the other ancient structures the program borrows as priors: the *method of loci* (spatial anchors improve recall — here, sector and radial coordinates serve as the loci); *memory palaces* (structured traversal over stored tokens — the evidence path is the palace walk, and it is returned per retrieval for audit); and *commentary traditions* layered summaries (gist nodes over raw leaves, i.e., the consolidation pyramid). What makes this more than decoration is that every prior is instantiated as an ablatable term: switch off radial, path, or gist scoring and the analogical structure’s contribution is a number, not a narrative. Ancient mnemonics suggested the coordinates; modern ablations decide whether they earn their weight — and on the measured record so far, radial is the smallest of the three structural contributors ($\Delta -0.0098$, $p=0.052$) while structural/activation carry the load. The analogy is welcome; the ledger is the science.

14.1 Grantha-grounded mechanisms, implemented (`core/indic_methods.py`)

Three further mappings from the granthas now exist as code, each traceable to a primary text, each [ENG] with its value claims (if any) kept [HYP], none affecting retrieval scores or the default-off guarantee:

1. **Pramāṇa provenance labels (Nyāya Sūtras).** Version-chain records carry a derived `pramana` label classifying *how* a fact is known: *pratyakṣa* (direct first-party input), *anumāna* (system-derived: consolidation, merge, gist), *upamāna* (comparison/analogy), *śabda* (testimony: documents, web,

citations) — with śabda as the honest default for unclassified sources. Nyāya’s point that these are *independent* knowledge sources is the groundwork for provenance-aware contradiction scoring (a śabda-śabda conflict is resolvable by newer testimony; a śabda-pratyakṣa conflict is not — future protocol work).

2. **Kaṭapayādi encoding (medieval Kerala mathematics).** The text→numeral table (with its reversal convention) is implemented as a deterministic secondary index for version chains: every returned record carries digits derived from the subject at read time — zero storage, unit-tested against the table. A memorable, checkable key in the tradition’s own sense.
3. **Śruti/smr̥ti preservation semantics (Mīmāṃsā).** This names an existing property rather than adding one: version chains are śruti-shaped (append-only, hash-linked, never rewritten; rollback appends and cites the superseded hash), while the working store is smr̥ti-shaped (decay, replay, consolidation may revise it). The śruti property is enforced by unit test.

Tests: `tests/test_indic.py` (4 checks: table values, classification, annotation purity, SQLite-restart annotation).

Reproducibility statement

Every measured number in this document is generated from committed run artifacts and reproduced by `paper/export_results.py --real`; the forgetting-lab table traces to `data_lab/forgetting_results.json` (variants A–K, 6 tasks, 30 questions each, complete); the retrieval tables trace to `data_bench/*.json`; consolidation and aging to `data_bench/consolidation_results.json` and `data_bench/aging_results.json`. Nothing here is hand-entered. Hypotheses H1–H6 are forward-looking and marked as such. As of this revision the constellation term (μ), the `VersionChain` store, the Born-rule backend (`ncm.backend: born`, H6’s mechanism verified as a unit property, benchmark value not yet evaluated), and the Python SDK (`mira_sdk.MemoryEngine: add / retrieve / explain / ask / remember / history / rollback / consolidate` over the thread-safe engine, explanations carrying all eleven components) are implemented; version chains are **SQLite-persisted** (append-only `versions` table; rehydration on restart with the cap re-applied; covered by `test_version_chain_sqlite_persistence`). The first exploratory **NCM pilot** (H2/H6 data points) is recorded in `data_bench/ncm_pilot_results.json` and §5.1 below; it is a pilot, not a confirmatory H2/H6 test. The pre-registered protocol lives at `docs/EXPERIMENT_PROTOCOL.md`, the audit at `docs/ARCHITECTURE_AUDIT.md`, and prior-art bounds (now including the 2026 landscape) at `docs/RELATED_WORK.md`.

References (selected)

1. Trivedi, H. et al. (2022). MuSiQue: Multihop Questions via Single-hop Question Composition. *TACL*, 10, 657–680.
2. Yang, Z. et al. (2018). HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. *EMNLP*.
3. McCloskey, M. & Cohen, N. (1989). Neural networks and catastrophic forgetting. *Neural Computation*, 11(7), 1649–1671.
4. Parisi, T. I. et al. (2019). Continual lifelong learning with neural networks: A review. *Neural Networks*, 113, 54–71.

5. Kirkpatrick, J. et al. (2019). Overcoming catastrophic forgetting in neural networks. *PNAS*, 116(13), 6521–6529.
6. Lopez-Paz, D. & Ranzato, M. (2017). Gradient Episodic Memory for Continual Learning. *NeurIPS*.
7. De Lange, M. et al. (2022). A continual learning survey. *IEEE TPAMI*, 44(7), 3366–3385.
8. McClelland, J. L., McNaughton, B. L., & O'Reilly, R. C. (1995). Complementary learning systems. *Psychological Review*, 102(3), 419–457.
9. Wilson, M. A. & McNaughton, B. L. (1994). Reactivation of hippocampal ensemble memories during sleep. *Science*, 265, 676–679.
10. Marr, D. (1971). Simple memory: A theory for archicortex. *Phil. Trans. R. Soc. B*, 262, 23–81.
11. Hebb, D. O. (1949). *The Organization of Behavior*. Wiley.
12. Bi, G.-Q. & Poo, M.-M. (1998). Activity-driven Hebbian modification of synaptic strengths. *J. Neuroscience*, 18, 10464–10472.
13. Robertson, S. & Zaragoza, H. (2009). The Probabilistic Relevance Framework: BM25 and beyond. *Foundations and Trends in IR*, 3(4).
14. Kanerva, P. (1988). *Sparse Distributed Memory*. MIT Press.
15. Park, J. S. et al. (2023). Generative Agents: Interactive Simulacra of Human Behavior. *UIST*.
16. Gutiérrez, B. J. et al. (2024). HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models. *NeurIPS*.
17. Edge, D. et al. (2024). From Local to Global: A Graph RAG Approach to Query-Focused Summarization. arXiv:2404.16130.
18. Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS*.
19. Hu, Y. et al. (2025). Memory in the Age of AI Agents: A Survey. arXiv:2512.13564.
20. Anonymous (2026). LongMemEval-V2: Evaluating Long-Term Agent Memory. arXiv:2605.12493. [verify against primary source]
21. Anonymous (2026). FOREVER: Forgetting Curve-Inspired Memory Replay for LLM Continual Learning. arXiv:2601.03938. [verify against primary source]
22. Huynh, L. et al. (2026). Quantum-inspired machine learning: a survey. *Computer Science Review* (2017–2025 corpus). [verify volume/pages]
23. Chirkova, N. et al. (2024). LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory. arXiv:2410.10813.
24. Park, J. S. et al. (2023). Generative Agents secondary comparisons and 2026 agent-memory vendor systems (Mem0, Letta/MemGPT, Zep) — vendor-reported numbers are cited as such in §12 and were not independently verified.